

COL1000: Lab 3

Semester 2, 2025–26

Instructions Upload your code to Gradescope as: `<entryNumber>_q<quesNumber>.py`. After uploading, run the autograder to check that all tests pass. Make sure not to include any custom prompt text in `input()` calls, and any extra text in `print()` statements. You must not use list conversion, or built-in operations such as `string-slicing`, `len`, `math.gcd()`, etc.

Question 1: Digit counting Write a program that reads a positive integer `n` and prints the number of even digits in it. For example, if the input is 9607004, then the output must be 5.

Question 2: Pattern printing Write a program that reads a positive digit `n` and using nested `for` loops prints an $n \times n$ matrix (without spaces) whose $(i, j)^{th}$ cell stores $\min(i, j)$. For example, if `n = 6`, the output should be:

```
1 1 1 1 1 1
1 2 2 2 2 2
1 2 3 3 3 3
1 2 3 4 4 4
1 2 3 4 5 5
1 2 3 4 5 6
```

Question 3: Co-prime pairs Write a program that reads a positive integer `n`, and using nested `for` loops and `break` counts the number of ordered pairs (i, j) such that:

$$1 \leq i < j \leq n \quad \text{and} \quad \gcd(i, j) = 1$$

For example, if `n = 5`, the output should be 9 as the coprime pairs are: $(1, 2)$, $(1, 3)$, $(1, 4)$, $(1, 5)$, $(2, 3)$, $(2, 5)$, $(3, 4)$, $(3, 5)$, and $(4, 5)$.

Question 4: Kaprekar numbers A number `n` is called a Kaprekar number if the square of `n` can be split into two non-negative integers `a` and `b` (not necessarily having the same number of digits) such that their sum equals `n`. For example, 45 is a Kaprekar number because $45^2 = 2025$ and splitting this as 20 and 25 gives $20 + 25 = 45$. Similarly, 4879 and 10 are Kaprekar numbers since their squares, 23804641 and 100, can be split into $(238, , 04641)$ and $(10, , 0)$ respectively, whose sums equal the original numbers.

Write a Python program that reads a positive integer `n` and prints `True` if it is a Kaprekar number, and `False` otherwise.